Would Either Shop Follow the Worst-Welfare Script?

Exact individual incentives under a recurring cap-and-guard rule

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1 A permitted adverse path is not automatically selfish behavior

Earlier stages identify the lowest customer benefit attainable by any allowed price choices. The graphical two-shop game deliberately lets a coordinator explore that adverse path. Neither fact establishes that separately profit-seeking shops would follow it. We now ask a narrower, explicit question: when the other shop always executes its highest permitted quote, can one shop earn more by choosing a different allowed sequence?

The same externally enforced cap-and-guard rule is retained. A real cap replaces the guard every K rounds; the first cap earns an actual reward. Between caps, each shop's allowed set uses the other shop's previous public price. The opponent remains a public deterministic script after every deviation. It does not learn, retaliate beyond that public rule, or access the focal shop's current unexecuted proposal.

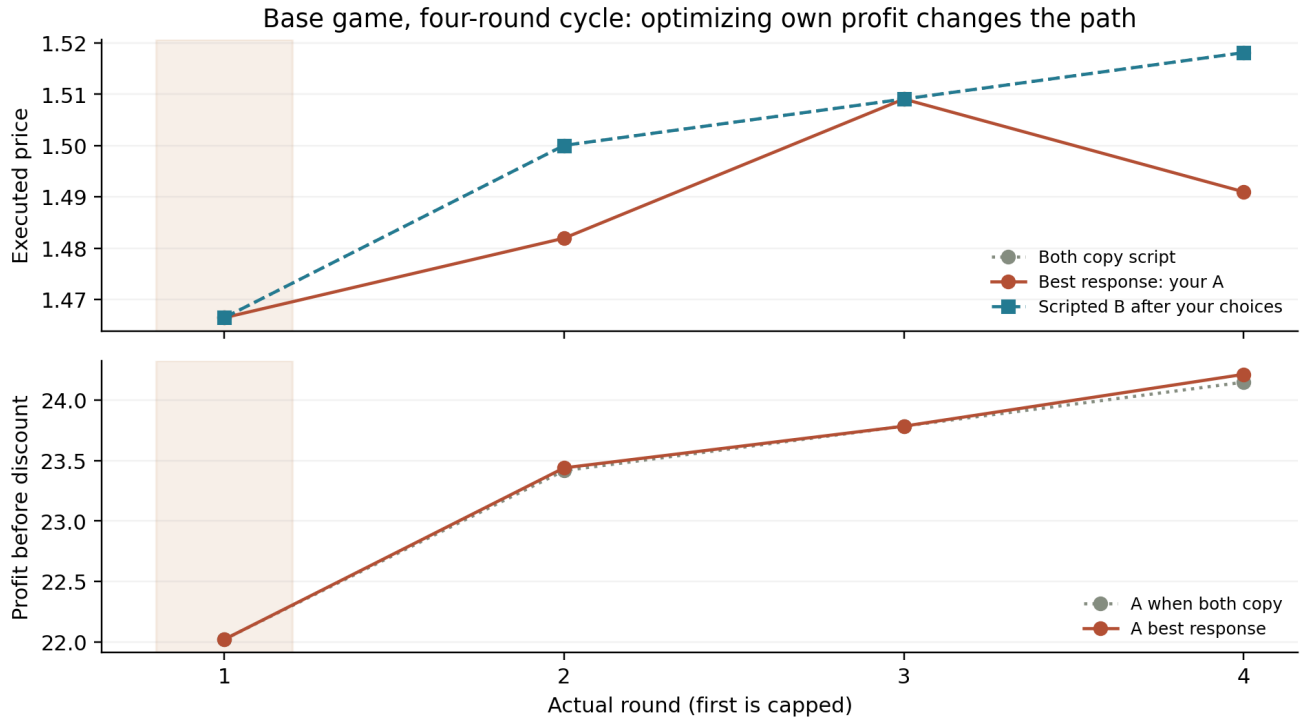


Figure 1. The separately audited base finite game, K=4, focal seller A, discount 19/20. Gray: both shops copy the highest-permitted-price script. Orange: A optimizes its whole block. Blue: B still uses the public guard rule after A changes its choices. In this example its displayed quote path happens to equal the gray baseline exactly, so the two lines overlap. All plotted prices and profits are direct saved-table lookups. Private profit and customer benefit are different outcomes.

Small, exact difference. Copying the script earns approximately 86.438659 in one discounted block; the best permitted strategy earns 86.516111, a gain of approximately 0.0896%. The same relative gain holds for repeated blocks. This is a useful stability caveat, not a large economic effect or autonomous collusion result.

2 Exact finite-block optimization, with a public opponent

Each inherited numerical game has n=61 or 62 saved price quotes including its original physical cap exactly. Let S(j) be the saved legal own-quote indices against previous rival index j, U(j)=max S(j), and A the cap index. The focal state (i,j) lists its own and the opponent's previous quote indices. In a guard phase the opponent chooses b=U(i), the focal action is a in S(j), and the next focal state is (a,b). Seller A reward is the saved table profitA[a*n+b]; seller B reward is profitB[b*n+a]. We compute both sellers independently; symmetry is not assumed by the algorithm.

For discount gamma=p/q in [0,1), let V_l be the largest discounted focal reward from guard phase l to the next cap, with V_K=0. The cap is phase 0, has opponent quote A, and allows all own indices through A:

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V_l(i,j) = max_(a in S(j)) { R(a,U(i)) + gamma*V_(l+1)(a,U(i)) }
V_0 = max_(a<=A) { R(a,A) + gamma*V_1(a,A) }
J_0 = V_0 / (1-gamma^K)
J_l(i,j) = V_l(i,j) + gamma^(K-l)*J_0
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These finite-horizon Bellman maxima cover every possible focal state and allowed action. A deterministic policy attains them; conditioning on more history or randomization cannot improve a finite-block maximum. The real cap makes the next block independent of its incoming history. Thus dividing by 1-gamma^K is justified for repeated blocks against this fixed opponent. For K=1, V_1=0 and the cap problem is immediate.

The script baseline has its own block value B_0 and continuation B_0/(1-gamma^K). Optimal and baseline continuations generally differ; they must not be canceled in phase comparisons. Every baseline-path phase is retained, including phases with zero gain. At gamma=0 the initial cap ignores all later rewards; a zero activation gain therefore need not mean zero later-phase gain.

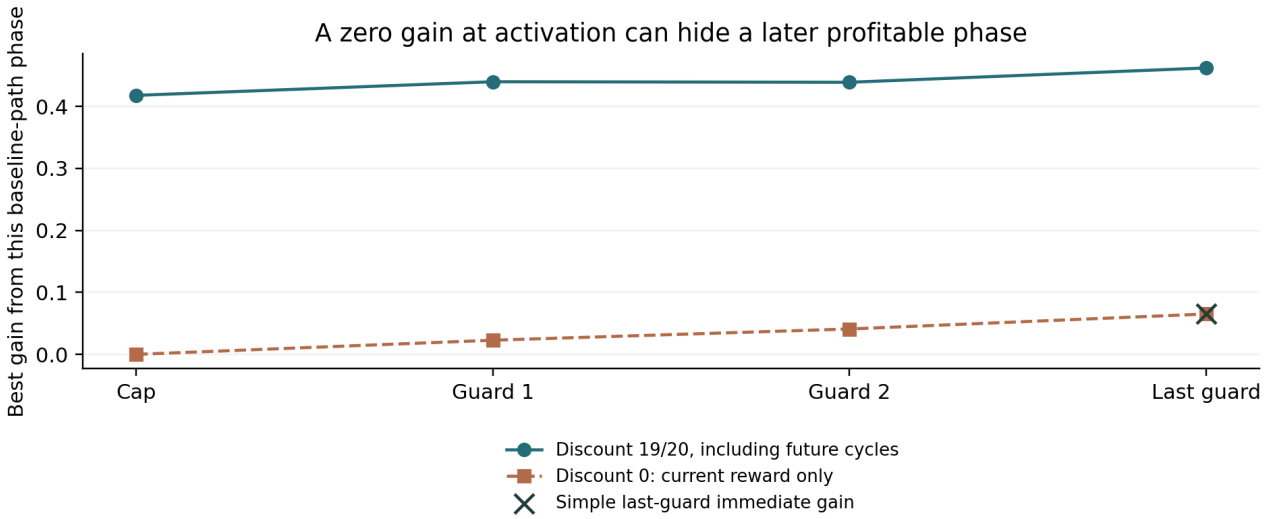


Figure 2. Base K=4, seller A, baseline-path phases. Discount 19/20 values include all future cycles; discount 0 values concern the current phase only. The cross is the elementary immediate gain in the last guard round. This is phase bookkeeping, not evidence of learning or a time-series experiment.

All saved binary64 profits are rational dyadic weights. Multiplying by a common power-of-two denominator and suitable q powers makes every Bellman comparison an exact integer comparison. Ties select the smallest original quote index. JavaScript BigInt implements the primary calculation; independent Python int/Fraction reconstructs all maxima without importing the primary backend. Display decimals never determine a sign or policy.

3 The simple explanation comes before the full calculation

At the last guard phase, the other shop's current price is fixed by completed history. A focal shop can choose a different permitted quote that increases its immediate profit. In the next real cap, both allowed sets reset; returning to the original cap quotes restores the same future baseline. Therefore a strictly positive last-guard one-shot gain already rejects the script as a best response from that phase for every 0<=gamma<1. From initial cap it remains positive for gamma>0, multiplied by gamma^(K-1). At gamma=0 that later gain does not affect the initial-cap value.

This elementary screen is positive for each seller in all eight inherited games at both recurring period slots. It explains every qualitative positive on-path instability found here; dynamic programming quantifies the best improvement rather than discovering an otherwise hidden mechanism. Permanent-cap K=1 uses an analogous immediate cap screen.

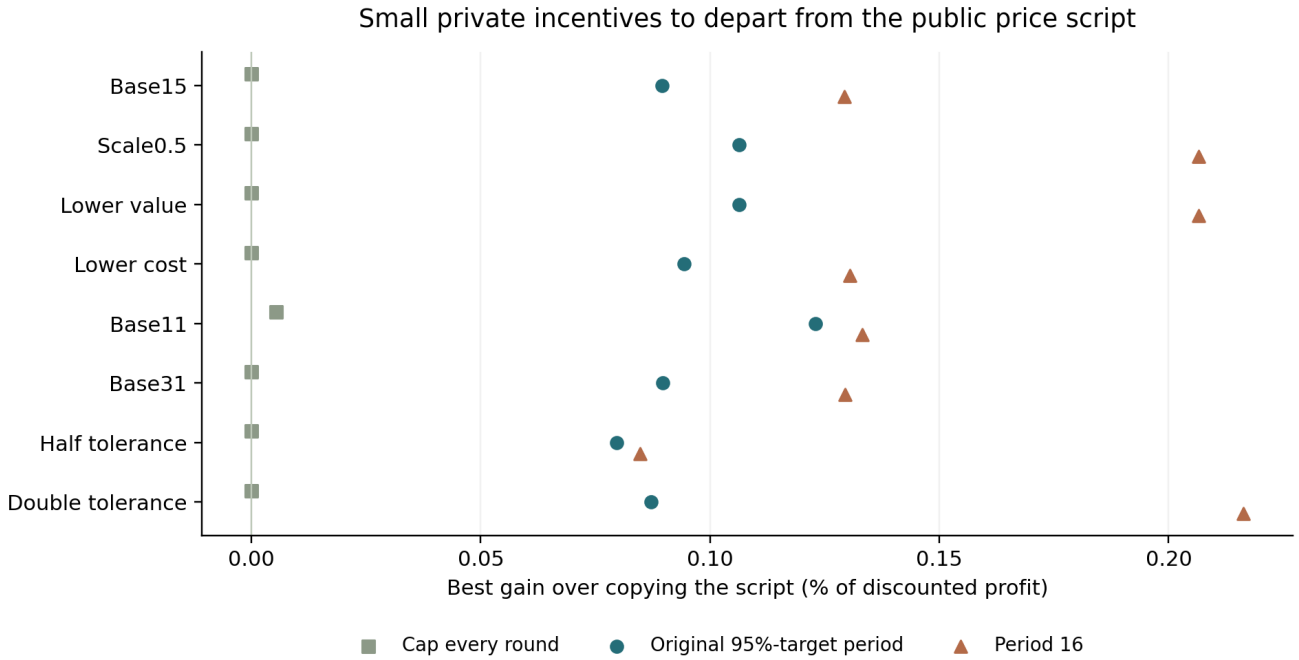


Figure 3. Seller A at gamma=19/20; both sellers are independently retained in the evidence. The original 95%-target period varies by inherited label; the fixed comparison is K=16. Zero permanent-cap points remain visible. Positive recurring gains are approximately 0.0797-0.2163% of baseline discounted profit. Original grid/tolerance labels and dimensionally equivalent markets are not independent replications.

The prospective matrix is eight saved games x three period slots x two focal sellers x two discounts = 96 retained labels. Slots are permanent cap K=1, the previously closed effective 95%-target period, and fixed K=16. Discounts are 19/20 and 0. All labels completed; none was dropped. Activation gains are 40 positive and 56 zero; 68 labels have a positive recorded phase gain, exactly matching 68 positive simple screens. At discount 19/20 all 32 recurring seller/slot labels have positive activation gain; at discount 0 there are 28 labels with zero cap gain but positive later-phase gain.

For permanent cap, seven of eight games have zero gain for both sellers; the original 11-price cap label has a tiny positive gain, about 0.00533%, reflecting the inherited physical cap. No new Nash cap was substituted. All 24 distinct period/game welfare certificates were reconstructed independently from saved rewards. All eight imported 95%-target periods meet their own finite-game relaxed 95% target; K=16 meets it in only three. This was checked explicitly, not transferred from the continuous period calculation. Every permitted deviation remains covered by its own undiscounted customer-benefit activation floor.

4 Verification and the limits of the claim

The protocol, theory, primary sources and transitive old inputs were frozen before any new 96-case execution. Fifteen primary synthetic tests cover asymmetric and negative rewards, exact ties, phase/discount/reset behavior, budgets and partial failures. Five independent tests include a separately generated asymmetric case and corrupted evidence. The primary run completed in 8.96 s; the largest after-record RSS sample was 153.24 MB, not a continuously measured peak. Primary numerical outputs were 5.73 MB before the completion record. The separate derived playable JSON is 11.22 MB and is outside that numerical-run output measurement.

The independent audit verifies 604 full guard layers, 2,302,192 state/cap Bellman maxima, 18,869,440 permitted-action comparisons, all 700 baseline-path phase diagnostics, every saved policy byte and smallest-index tie choice, full optimal-block replays, 24 welfare certificates and 30,506 saved payoff/mask pairs. Six actual evidence corruptions are rejected. These are verification-coverage counts, not independent experimental trials. Integer arithmetic backends and language implementations differ; their shared mathematical formulation remains a possible common modeling error.

The optional human profit challenge uses the same legal masks and exact weighted ledger, one public opponent and one actual cap-first block. It reveals stored optimal choices only when requested; it performs no browser DP or model call. No shortfall or win/lose classification is assigned until the complete matching block. The game locks on missing evidence; invalid stored actions and inconsistent completed scores are rejected before accepting the affected action. Developer control checks are not human play observations or a usability study.

Closest prior work and contribution

Unilateral policy auditing and dynamic programming are established methods. Lopes de Moraes and De Losso (2026), *Algorithmic Collusion Without Deterrence: Auditing the Punishment Mechanism in Q-Learning Pricing Agents*, arXiv:2610.21867v2, is especially close: it audits learned pricing policies and unilateral deviations. Calvano et al. (2020), AER 110, 3267-3297, doi:10.1257/aer.20190623, studies learning and collusion in pricing. Our exact DP is not a new algorithm, the public upper script is not a learned policy, and we do not claim priority for incentive audits, price-grid checks, resets or shielding. The limited contribution is a fully reproducible incentive check separating this controller's all-policy welfare envelope from one particular script's individual attractiveness.

Limitations, dual use and ethics

A positive gain rejects only the specified script against a fixed public opponent at the stated phase and discount. A zero on-path gain is not subgame perfection, general equilibrium, robustness to other opponents or learned stability. The calculation concerns saved numerical weights and finite masks, not an exact continuous-game incentive certificate. No market-estimated parameters, behavior of independent adaptive agents, experiments with LLMs, real firms or human participants are provided. The small sizes could make the differences unimportant to particular noisy learners; that behavioral question remains open.

The toy shop model and explicit protection rule are intended for evaluating safeguards, and should not be presented as a strategy for coordinating real firms. There is no participant data collection, communication among real sellers, paid API use, model/account game player, external deployment or submission in this stage. Codex and AI agents substantially assisted formulation, implementation, independent AI-agent auditing, analysis and writing under Mohammadpayam Teabi's direction. Independent-agent review is not human peer review.

Reproduce read-only: node --test --test-isolation=none tests/incentive-study.test.js tests/incentive-game.test.js; python -m unittest discover -s tests -p test_incentive_audit.py; python research/audit-incentive-study.py. Descriptive rebuilds: node research/incentive-analysis.js; node research/build-incentive-web.js; python research/build-incentive-figures.py; python research/build-incentive-note.py. Frozen hashes, all 96 records, journals, failures, layer hashes, policy bytes and all zero results are retained.

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